

PROBLEM STATEMENT #11 | Healthcare & Telemedicine

Telemedicine Slot Booking & Prescription Portal

1. Problem Context & Overview

A secure healthcare consultation portal allowing remote patients to view doctor specialties, book video consultation slots, receive encrypted tele-consultation room links, and download digital signed prescriptions.

Target Stakeholders / Actors: *Patient, Attending Physician*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.

Sample Acceptance Criteria: Pass: Valid cryptographic signature attached to prescription PDF; Fail: Prescription generated without doctor license validation.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: All video session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest with AES-256.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.